AR 2EB

ALAMEDA RESEARCH
2: ELECTRIC BOOGALOO
LONG HORIZONS · STRUCTURAL SHIFTS · IMAGINATION

INTERNAL RESEARCH · MEMO · NOT INVESTMENT ADVICE · AI-ASSISTED

Archer Aviation

NYSE : ACHR · Young-Company DCF (Damodaran)

$5.4B mkt cap · 790M sh · $1.78B cash, $80M net debt · $1.78B cash ($0.95B + $0.83B ST inv), low debt; ~760M basic / ~790M diluted; Pre-revenue (Q1'26 first $1.6M); ~$727M/yr burn, ~2.4yr runway; Anduril defense JV + $142M USAF; ~$6B order book (mostly LOIs)

31 May 2026 · the qualitative

$6.81

THE CENTRAL QUESTION

Is Archer's ~half-Joby price enough discount to make the eVTOL-plus-defense lottery ticket positive-EV — despite being the certification follower?

Archer trades at ~$5.4B (≈$6.81/sh) on $0 revenue — about half Joby's cap on a near-identical certification position, with $1.78B cash but a ~$727M/yr burn (≈2.4yr runway) and a defense wildcard (the Anduril autonomous-VTOL JV) Joby lacks. The young-company DCF asks what aircraft fleet, UAM-plus-defense revenue, and dilution are plausible over a decade. The finding: 85% of the probability sits well below spot — heavy dilution craters the bear/base outcomes — but Archer is cheap enough that the 15% cert-success-plus-defense tail (paying 4-13×) lifts the weighted expected modestly above today's price.

PROBABILITY-WEIGHTED EXPECTED VALUE

$8.01 expected fair value today

+17.7% vs spot $6.81

FORWARD COMPOUNDED VALUE

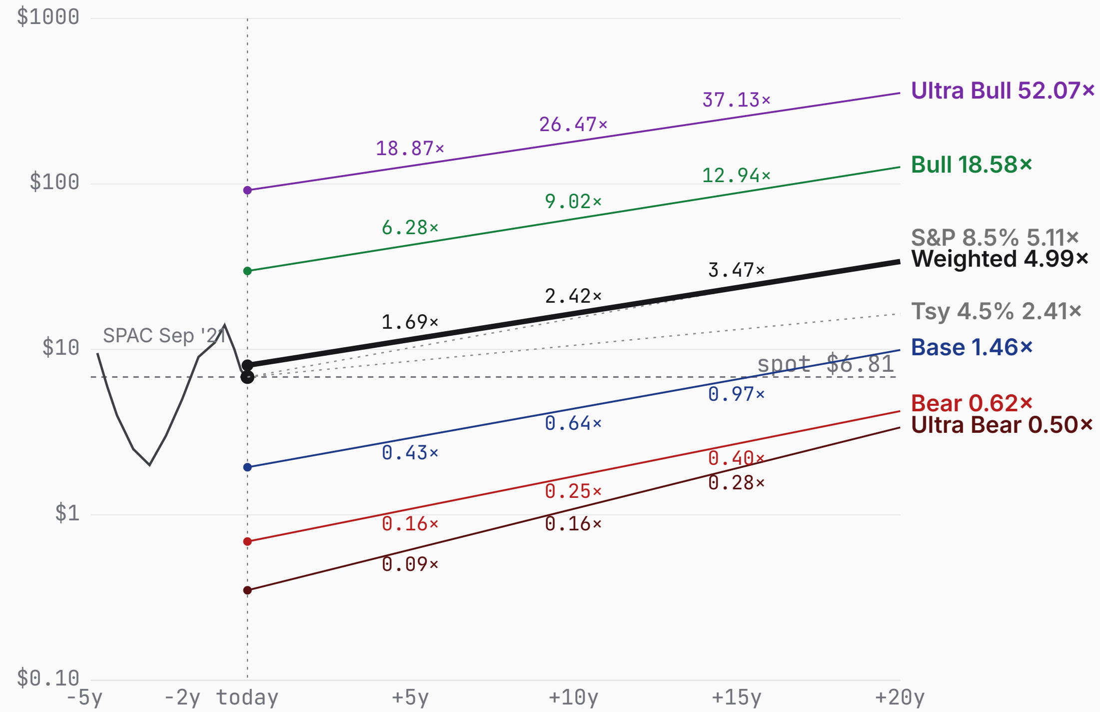
+5y	+10y	+15y	+20y
$11.48	$16.47	$23.64	$33.99
1.69× spot	2.42× spot	3.47× spot	4.99× spot

WEIGHTING RATIONALE

Ultra Bear 12% TC fails (Lilium-style); cash burns; near-total loss.
Bear 31% Cert slips; lean sub-scale operator; dilution wins.
Base 42% Cert ~2027; mid-scale UAM + defense; diluted.
Bull 11% Cert + DoD program of record; ~5× spot.
Ultra Bull 4% UAM leader + major defense program; ~14×.

Bull (11%) + Ultra Bull (4%) together contribute $6.94 — 87% of the $8.01 expected value despite 15% combined probability. Today's spot ($6.81) sits 15% below the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR	BEAR	BASE	BULL	ULTRA BULL
$0.35	$0.69	$1.94	$29.79	$91.64
-94.9% vs spot	-89.9% vs spot	-71.5% vs spot	+337.4% vs spot	+1245.7% vs spot
TAM 0.0% P(fail) 45% S2C 0.6 Prob 12%	TAM 0.5% P(fail) 35% S2C 1.0 Prob 31%	TAM 2.2% P(fail) 18% S2C 1.2 Prob 42%	TAM 6.4% P(fail) 8% S2C 1.5 Prob 11%	TAM 9.6% P(fail) 4% S2C 1.7 Prob 4%
TC fails (Lilium-style); cash burns; dilution.	Cert slips; lean sub-scale operator.	Cert ~2027; UAM scales + defense kicker.	Cert + DoD program of record win.	UAM leader + major defense program.

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Archer Aviation scenario narratives

PROBABILITY WEIGHTS

Ultra Bear 12% Bear 31% Base 42% Bull 11% Ultra Bull 4% · Weighted expected $8.01 (+17.7% vs spot)

ULTRA BEAR $0.35

-94.9% vs spot

Probability 12%

TC fails (Lilium-style); cash burns; dilution.

WHY 12%

The category's base rate is unforgiving — no one holds an FAA Type Certificate yet, and Lilium (raised $1.4B, failed) proves under-capitalization is fatal. Archer is the cert-follower with ≈2.4yr of runway; 45% within-scenario failure reflects real first-of-a-kind certification + funding risk. 12% scenario weight.

WHAT HAPPENS

Type certification defeats Archer — a flight-test setback, a transition-flight failure, or simply running out of runway (≈2.4yr of cash at the ~$727M/yr burn) before the FAA grants a TC. Lilium's 2025 collapse proves the category kills the under-capitalized. Archer dilutes desperately at collapsing prices, then restructures; the residual is defense scraps plus whatever cash survives the attempt.

With a 45% within-scenario failure probability and the share count ballooning past 1.4B on sub-$4 raises, the expected per-share collapses toward the distress floor — a near-total loss from today's $6.81.

BEAR $0.69

-89.9% vs spot

Probability 31%

Cert slips; lean sub-scale operator.

WHY 31%

Cert slips and the business stays sub-scale; the ~$6B order book (mostly LOIs) converts only partially. 35% failure probability — Archer is behind Joby on conforming-aircraft flight testing. 31% weight as a very plausible 'certifies but the dilution wins' outcome.

WHAT HAPPENS

Archer eventually certifies but late and sub-scale; it becomes a lean niche eVTOL operator (~$1.2B revenue by the mid-2030s at a low-teens margin) plus a modest defense line, never converting the ~$6B order book or matching Joby's cadence. Heavy dilution (≈$2.4B of raises at depressed prices) funds the long road to cert.

The DCF lands under $1/share — the market is pricing a far larger outcome. This is the 'cert works, but the dilution math wins' world that Joby's bear shares.

BASE $1.94

-71.5% vs spot

Probability 42%

Cert ~2027; UAM scales + defense kicker.

WHY 42%

Requires cert on roughly schedule and a mid-scale UAM-plus-defense ramp — no category dominance. 18% failure probability given $1.78B cash, Stellantis manufacturing, and the USAF/Anduril defense anchor. 42% weight as the central outcome.

WHAT HAPPENS

The modal path: Archer certifies on roughly the targeted timeline (~2027), scales the Midnight UAM network across the US and UAE, and converts part of the Anduril defense relationship into recurring revenue — ~$5.6B combined by 2035 at a low-teens operating margin. A real, profitable transport-tech business.

But ~$3.4B of dilutive raises to fund the buildout balloon the share count past 1.2B, and most of the fleet's value sits beyond the 10-year window. DCF ~$2.25/share — ~67% below spot, the dilution tax on a successful-but-capital-hungry ramp.

BULL $29.79

+337.4% vs spot

Probability 11%

Cert + DoD program of record win.

WHY 11%

A chain: cert clears AND manufacturing ramps AND UAM scales AND the Anduril relationship becomes a DoD program of record. Each plausible at 50-70%; jointly ~11%. The defense leg is Archer's distinctive second shot at the upside.

WHAT HAPPENS

The bull: Archer certifies, the Covington/Stellantis line ramps toward high volume, the UAM network scales across multiple metros, AND the Anduril partnership converts into a DoD program of record — ~$16B combined revenue by 2035 at a ~28% operating margin. The defense leg is the differentiator Joby lacks.

Self-funding arrives mid-decade; dilution turns value-accretive. DCF ~$32/share — ~5× today's price. This is the scenario the cheap entry is really buying.

ULTRA BULL $91.64

+1245.7% vs spot

Probability 4%

UAM leader + major defense program.

WHY 4%

Tail of tails: UAM leadership AND a major multi-service defense program AND a premium platform terminal. Individually 40-60%; jointly ~4%. The asymmetric upside that the half-Joby price is really buying.

WHAT HAPPENS

The tail: Archer becomes a US UAM leader AND lands a major multi-service defense program — the autonomous hybrid-VTOL scales into a real DoD platform — for ~$24B combined revenue by 2035 at a ~34% operating margin and a premium platform terminal.

DCF ~$95/share — ~14× spot. A genuine 'eVTOL + defense prime' outcome; each leg is individually plausible at 40-60%, jointly ~4%. The asymmetric upside that makes the cheap entry positive-EV.

INTERNAL RESEARCH · MEMO · NOT INVESTMENT ADVICE · AI-ASSISTED

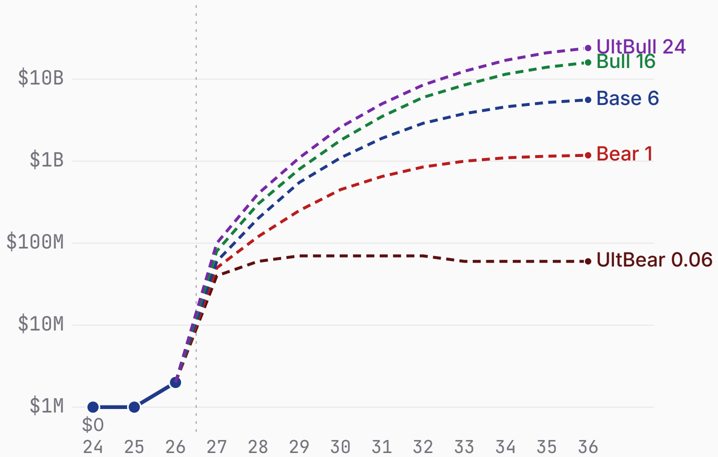
31 May 2026 · the snapshot

Archer Aviation business snapshot

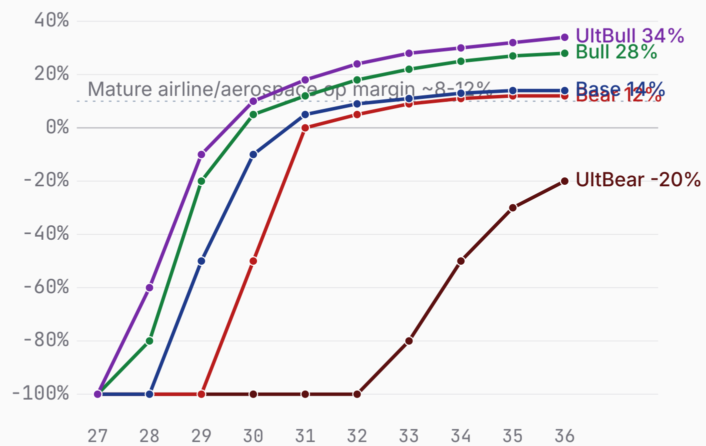
Bear $0.69 Base $1.94 Bull $29.79

FY26-FY35 scenario projections (pre-revenue today) · fiscal years end Dec 31 · Q1 2026 10-Q, FY25 results

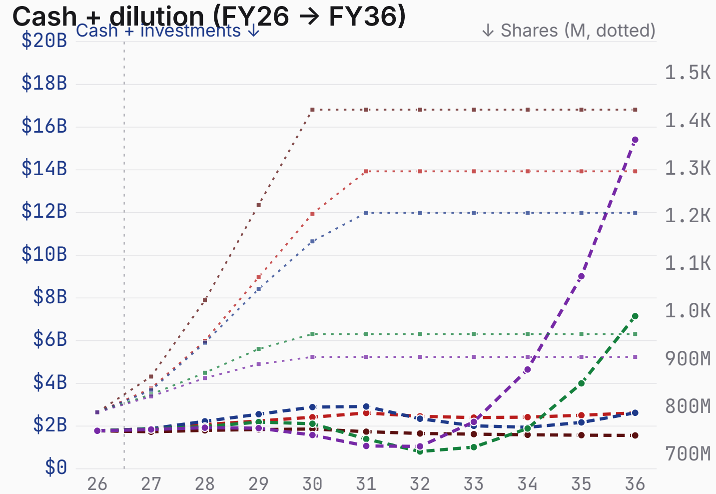
Revenue — history + scenarios to FY36 (log)



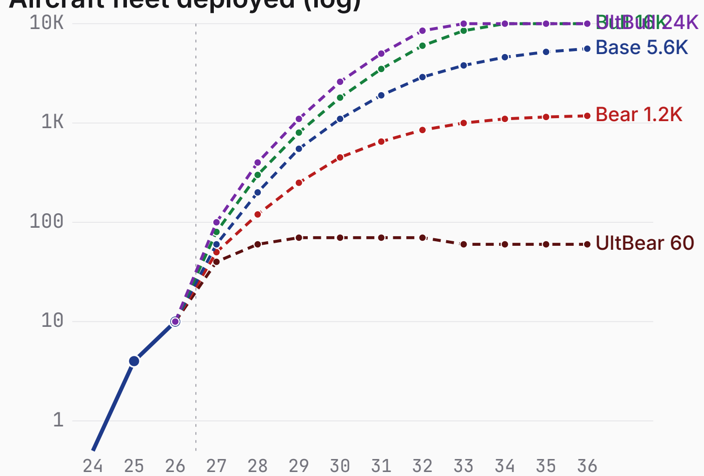
Path to profitability — op margin (FY27→FY36)



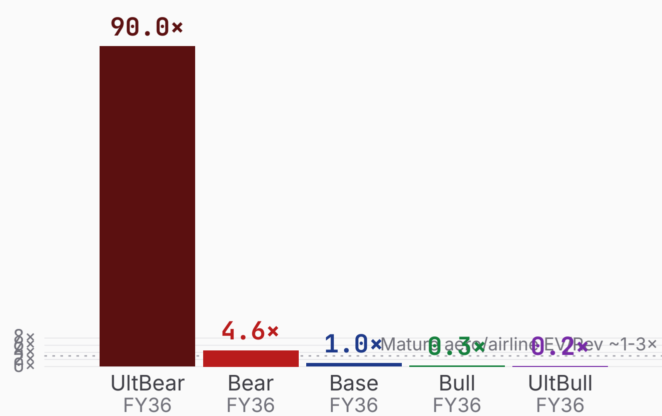
Cash + dilution (FY26 → FY36)



Aircraft fleet deployed (log)

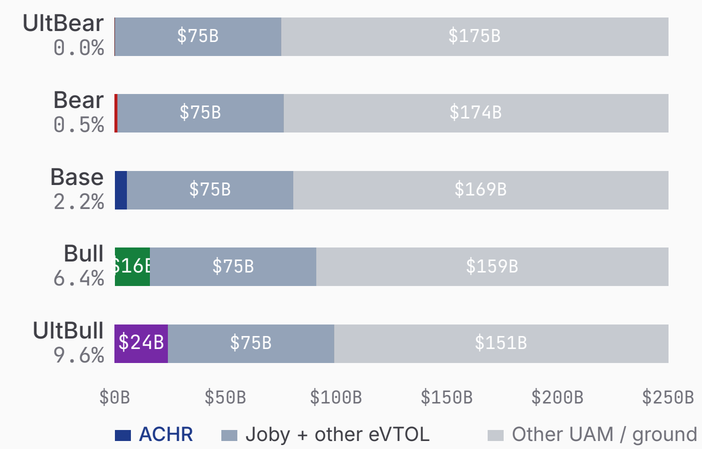


$5.40B mkt cap as P/S on FY36 rev



On $0 revenue today, every scenario implies the multiple resolves only as UAM +

$250B FY36 global UAM TAM — ACHR share (+ separate



Sources: Archer Q1 2026 10-Q (cash+ST inv $1.78B, net loss), FY2025 results (net loss $618M), order-book + Anduril/USAF disclosures. Revenue modeled as UAM (passenger) + defense; FCF is NOL-shielded pre-tax over the explicit window (documented simplification). TAM: $250B FY36 global UAM (matches the Joby memo) plus a separate defense leg.

Archer Aviation 7 Powers · the moat, scored

Bear $0.69 Base $1.94 Bull $29.79

Arena. Electric vertical-takeoff air mobility (eVTOL) + defense VTOL — Archer (Midnight UAM 'airline' + Anduril defense) vs Joby (cert + cash leader), Beta, Vertical, EHang (China), Wisk/Boeing (autonomy); the race to originate the first certified, scaled UAM + dual-use position.

POWER ORIGINATION
origination window: open

POWER ORIGINATION · 7 POWERS

Scale Economies	1/3	The Covington/Stellantis line targets high volume, but it's pre-cert and unproven at rate.
Network Economies	1/3	A vertiport/route-network flywheel is future, not yet originated.
Counter-Positioning	2/3	Archer's defense dual-use (Anduril) is a position Joby and pure-UAM peers can't easily mirror.
Switching Costs	1/3	Airline/operator + defense-program lock-in is real once contracted, but contracts are early/conditional.
Branding	2/3	A leading eVTOL brand (United, Stellantis, Anduril, UAE); strong but unproven on delivery.
Cornered Resource · thesis leans here	2/3	An FAA TC (once granted) is the first-mover regulatory moat; Archer is a close #2 with manufacturing + defense assets.
Process Power	1/3	No certified aircraft yet; the manufacturing learning curve is ahead, not demonstrated.

THE RACE · NAMED RIVALS

Joby Aviation — Toyota; ~$2.5B cash
~20% terminal share · Cert + cash leader (Toyota ~$894M, ~$2.5B cash, Dubai 6yr exclusive); flew FAA-conforming aircraft for TIA.

Beta Technologies — >$1B; QIA, GE
~12% terminal share · eCTOL flying/delivering; eVTOL behind; strong balance sheet (Nov'25 IPO ~$7.4B); cargo + military.

Vertical Aerospace — ~$180M; Mudrick
~6% terminal share · VX4; targets cert end-2026; chronically thin balance sheet.

EHang — China; ~$97M
~10% terminal share · EH216-S autonomous; CERTIFIED in China (tourism) — but jurisdiction-bound.

Wisk (Boeing) — Boeing $450M+
~10% terminal share · Autonomous (no pilot) Gen 6; harder/longer cert, later EIS; Boeing-funded.

LEAD / LAG · FALSIFIERS

FAA certification stage	closed 'Phase 3'; for-credit testing vs Joby (TIA, conforming a/c flown)	LAGGING
Defense / dual-use	Anduril JV + $142M USAF vs Joby (limited DoD)	LEADING
Liquidity / runway	$1.78B (~2.4yr) vs Joby ~$2.5B	LAGGING

COMPETITIVE READ

Archer is originating a credible #2 eVTOL position plus a distinctive defense dual-use leg (Anduril) that Joby lacks — at roughly half Joby's market cap. But it lags Joby on the one Power that matters first (FAA certification — Joby has flown conforming aircraft for TIA; Archer hasn't), and on cash/runway; the category just killed the under-capitalized Lilium. The window is open as the close #2 with a defense call option, but certification and the ~$727M/yr burn are the falsifiers the whole bull case must clear.

INTERNAL RESEARCH · MEMO · NOT INVESTMENT ADVICE · AI-ASSISTED

31 May 2026 · the quantitative

Archer Aviation show your work

Bear $0.69 Base $1.94 Bull $29.79 · Weighted $8.01 (+17.7%)

ULTRA BEAR $0.35

ASSUMPTIONS

TAM Year-10 ($B)	250
Mature share (%)	0.0
Mature op margin (%)	-20
Sales-to-capital	0.6
Terminal WACC (%)	12.0
Terminal growth (%)	2.0
P(failure) (%)	45
Scenario probability (%)	12

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.04	-700%	-0.35	-0.30
FY28	0.06	-500%	-0.33	-0.25
FY29	0.07	-350%	-0.26	-0.17
FY30	0.07	-250%	-0.17	-0.10
FY31	0.07	-180%	-0.13	-0.06
FY32	0.07	-120%	-0.08	-0.04
FY33	0.06	-80%	-0.03	-0.01
FY34	0.06	-50%	-0.03	-0.01
FY35	0.06	-30%	-0.02	-0.01
FY36	0.06	-20%	-0.01	-0.00

Σ PV explicit FCF	-0.94
+ PV terminal (g 2.0%)	-0.03

EQUITY BUILD · $B & PER SHARE

Operating EV	$-0.97B
+ Cash & investments	$1.78B
= Equity value	$0.73B
Raised over period	$1.20B
Dilution by FY36	+80%
FY36 shares (M)	1,425
DCF per share	$0.51
× (1−P_fail) [55%]	$0.28
+ P_fail × distress [45% × $0.15]	$0.07
= Expected per share	**$0.35**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$0.62	$1.09	$1.92	$3.38
0.09×	0.16×	0.28×	0.50×

BEAR $0.69

ASSUMPTIONS

TAM Year-10 ($B)	250
Mature share (%)	0.5
Mature op margin (%)	12
Sales-to-capital	1.0
Terminal WACC (%)	9.5
Terminal growth (%)	2.0
P(failure) (%)	35
Scenario probability (%)	31

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.05	-300%	-0.20	-0.17
FY28	0.12	-200%	-0.31	-0.24
FY29	0.25	-120%	-0.43	-0.29
FY30	0.45	-50%	-0.42	-0.26
FY31	0.65	+0%	-0.20	-0.11
FY32	0.85	+5%	-0.16	-0.08
FY33	1.00	+9%	-0.06	-0.03
FY34	1.10	+11%	+0.02	+0.01
FY35	1.15	+12%	+0.09	+0.03
FY36	1.18	+12%	+0.11	+0.04

Σ PV explicit FCF	-1.10
+ PV terminal (g 2.0%)	0.50

EQUITY BUILD · $B & PER SHARE

Operating EV	$-0.60B
+ Cash & investments	$1.78B
= Equity value	$1.10B
Raised over period	$2.40B
Dilution by FY36	+64%
FY36 shares (M)	1,296
DCF per share	$0.85
× (1−P_fail) [65%]	$0.55
+ P_fail × distress [35% × $0.40]	$0.14
= Expected per share	**$0.69**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$1.09	$1.71	$2.69	$4.24
0.16×	0.25×	0.40×	0.62×

BASE $1.94

ASSUMPTIONS

TAM Year-10 ($B)	250
Mature share (%)	2.2
Mature op margin (%)	14
Sales-to-capital	1.2
Terminal WACC (%)	8.5
Terminal growth (%)	2.5
P(failure) (%)	18
Scenario probability (%)	42

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.06	-250%	-0.20	-0.18
FY28	0.20	-120%	-0.36	-0.28
FY29	0.55	-50%	-0.57	-0.40
FY30	1.10	-10%	-0.57	-0.36
FY31	1.90	+5%	-0.57	-0.33
FY32	2.90	+9%	-0.57	-0.29
FY33	3.80	+11%	-0.33	-0.15
FY34	4.60	+13%	-0.07	-0.03
FY35	5.20	+14%	+0.23	+0.09
FY36	5.60	+14%	+0.45	+0.16

Σ PV explicit FCF	-1.76
+ PV terminal (g 2.5%)	2.78

EQUITY BUILD · $B & PER SHARE

Operating EV	$1.02B
+ Cash & investments	$1.78B
= Equity value	$2.72B
Raised over period	$3.40B
Dilution by FY36	+53%
FY36 shares (M)	1,209
DCF per share	$2.25
× (1−P_fail) [82%]	$1.85
+ P_fail × distress [18% × $0.50]	$0.09
= Expected per share	**$1.94**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$2.92	$4.39	$6.60	$9.92
0.43×	0.64×	0.97×	1.46×

BULL $29.79

ASSUMPTIONS

TAM Year-10 ($B)	250
Mature share (%)	6.4
Mature op margin (%)	28
Sales-to-capital	1.5
Terminal WACC (%)	7.5
Terminal growth (%)	3.0
P(failure) (%)	8
Scenario probability (%)	11

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.08	-200%	-0.21	-0.19
FY28	0.30	-80%	-0.39	-0.31
FY29	0.80	-20%	-0.49	-0.36
FY30	1.80	+5%	-0.58	-0.38
FY31	3.50	+12%	-0.71	-0.42
FY32	6.00	+18%	-0.59	-0.32
FY33	8.50	+22%	+0.20	+0.10
FY34	11.50	+25%	+0.88	+0.40
FY35	14.00	+27%	+2.11	+0.90
FY36	16.00	+28%	+3.15	+1.24

Σ PV explicit FCF	+0.67
+ PV terminal (g 3.0%)	28.43

EQUITY BUILD · $B & PER SHARE

Operating EV	$29.10B
+ Cash & investments	$1.78B
= Equity value	$30.80B
Raised over period	$2.00B
Dilution by FY36	+21%
FY36 shares (M)	954
DCF per share	$32.29
× (1−P_fail) [92%]	$29.71
+ P_fail × distress [8% × $1.00]	$0.08
= Expected per share	**$29.79**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$42.77	$61.40	$88.14	$126.54
6.28×	9.02×	12.94×	18.58×

ULTRA BULL $91.64

ASSUMPTIONS

TAM Year-10 ($B)	250
Mature share (%)	9.6
Mature op margin (%)	34
Sales-to-capital	1.7
Terminal WACC (%)	7.0
Terminal growth (%)	3.5
P(failure) (%)	4
Scenario probability (%)	4

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.10	-180%	-0.24	-0.21
FY28	0.40	-60%	-0.42	-0.34
FY29	1.10	-10%	-0.52	-0.39
FY30	2.60	+10%	-0.62	-0.42
FY31	5.00	+18%	-0.51	-0.32
FY32	8.50	+24%	-0.02	-0.01
FY33	12.50	+28%	+1.15	+0.60
FY34	17.00	+30%	+2.45	+1.19
FY35	21.00	+32%	+4.37	+1.98
FY36	24.00	+34%	+6.39	+2.70

Σ PV explicit FCF	+4.79
+ PV terminal (g 3.5%)	79.94

EQUITY BUILD · $B & PER SHARE

Operating EV	$84.73B
+ Cash & investments	$1.78B
= Equity value	$86.43B
Raised over period	$1.60B
Dilution by FY36	+15%
FY36 shares (M)	906
DCF per share	$95.40
× (1−P_fail) [96%]	$91.58
+ P_fail × distress [4% × $1.50]	$0.06
= Expected per share	**$91.64**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$128.53	$180.27	$252.84	$354.62
18.87×	26.47×	37.13×	52.07×

Archer Aviation supporting analysis · disclaimers · glossary

Bear $0.69 Base $1.94 Bull $29.79

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

1 **Half of Joby's price, similar cert position.**
~$5.4B cap vs Joby's ~$11B on a near-identical TC stage — the cheaper entry is the whole thesis.

2 **Defense optionality Joby lacks.**
Anduril autonomous-VTOL JV + $142M USAF AFWERX + EDGE/Omen powertrain — a second revenue leg and a DoD-program-of-record call option.

3 **Stellantis as contract manufacturer.**
Automotive mass-production discipline behind the Covington line; the manufacturing partner is hands-on, not just strategic.

4 **$1.78B cash, low debt.**
Funds the next phase; the constraint is burn (~$727M/yr), not insolvency today.

5 **First to close FAA 'Phase 3'.**
Real certification progress; the gap to Joby is conforming-aircraft flight testing, not paperwork.

FALSIFICATION TRIGGERS

Bull validation
FAA Type Certificate (2026-27) · Anduril → DoD program of record · UAE/US commercial ops generating revenue · Covington line at rate

Bear validation
TC slips beyond 2027 · order book stays LOIs · burn forces dilutive raises < $5/share

Reframe needed
if a major DoD program of record is awarded, the defense leg re-rates independently of UAM — revisit the bull/ultra weights

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GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Midnight — Archer's 4-passenger + pilot eVTOL, ~100-mile range, 12 tilt-rotors; targets short city hops.

Launch Edition — Archer's direct-operations model in international markets (UAE, etc.) ahead of US commercial UAM.

Type Certification (TC) — The FAA's airworthiness approval; no eVTOL has one yet — the gating risk for the whole sector.

Sales-to-capital (s2c) — Damodaran reinvestment efficiency — incremental revenue per dollar of reinvested capital.

Anduril JV — Archer's partnership to build an autonomous hybrid-VTOL for a potential DoD program of record — the defense leg Joby lacks.